

AP Chemistry Assignment: Analyzing Photoelectron Spectra

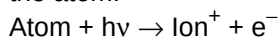
Topic: Photoelectron spectroscopy, electron configurations, ionization energy, and periodic trends

Suggested time: 45–60 minutes

Format: AP-style multiple-choice and free-response questions

Background

In photoelectron spectroscopy, atoms are exposed to high-energy photons. The photons eject electrons from the atom:



The kinetic energy of an emitted electron is related to the photon energy and the electron's binding energy:

$$\text{KE} = h\nu - E_{\text{B}}$$

For this assignment, assume that each peak represents electrons in one occupied subshell, peak area/intensity is proportional to electron count in that subshell, binding energy increases as electrons are held more strongly, and all spectra were obtained with photons of the same energy.

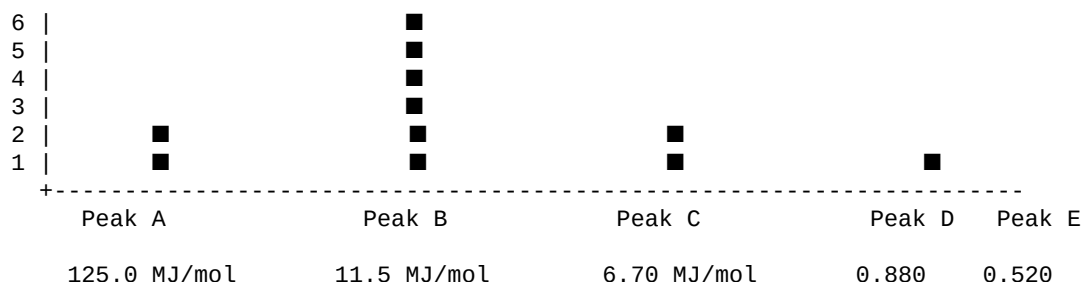
Part I: Interpreting a Photoelectron Spectrum

The photoelectron spectrum of an unknown element, **Element X**, is shown below.

In each diagram: horizontal position = binding energy; peak height = relative intensity/electron count; higher binding energy is farther left.

Binding energy increases ←  → Binding energy decreases

Relative intensity



Questions 1–6

- Which subshell is represented by Peak A? A. 1s B. 2s C. 2p D. 3s E. 3p
- Which subshell is represented by Peak E? A. 1s B. 2s C. 2p D. 3s E. 3p
- What is the total number of electrons in Element X? A. 11 B. 12 C. 13 D. 14 E. 15
- Which is the electron configuration of Element X? A. $1s^2 2s^2 2p^6 3s^2$ B. $1s^2 2s^2 2p^6 3s^1$ C. $1s^2 2s^2 2p^6 3s^2 3p^1$ D. $1s^2 2s^2 2p^6 3s^2 3p^2$ E. $1s^2 2s^2 2p^6 3s^1 3p^2$
- Which element is Element X most likely to be? A. Na B. Mg C. Al D. Si E. P
- Why is the binding energy of the 2p electrons greater than that of the 3s electrons? A. 3s electrons have greater mass. B. 2p electrons experience greater effective nuclear charge and are closer to the nucleus. C. 2p electrons occupy a higher principal level. D. 3s electrons are shielded by 2p electrons only. E. 2p electrons have greater kinetic energy because they are farther from the nucleus.

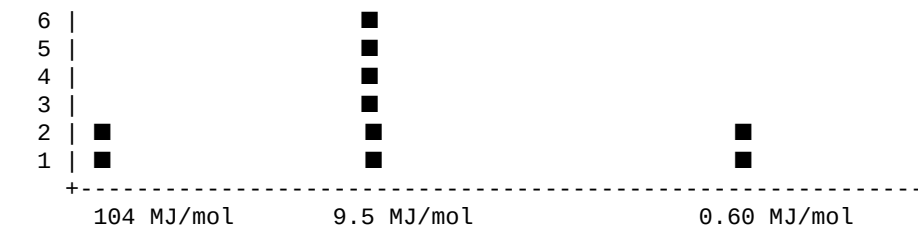
Part II: Comparing Spectra

The simplified photoelectron spectra for three elements are shown below. Each block represents approximately one electron in a subshell.

Element A

Binding energy increases ←  → Binding energy decreases

Relative
intensity

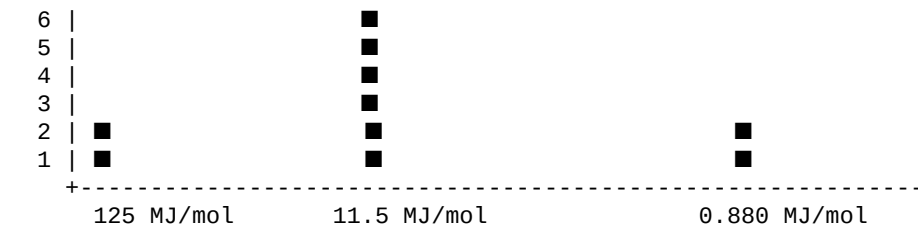


Electron distribution: $1s^2 2s^2 2p^4 3s^1$

Element B

Binding energy increases ←  → Binding energy decreases

Relative
intensity

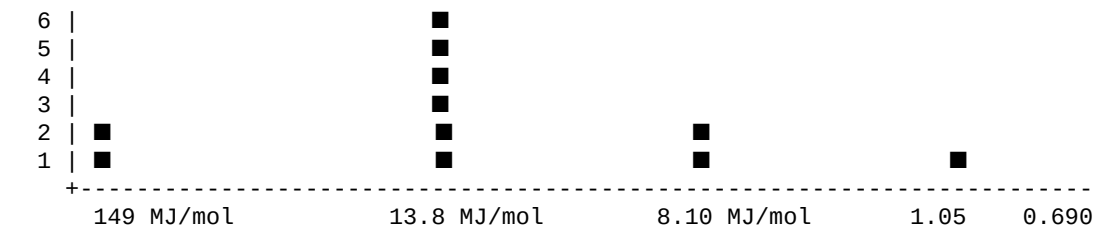


Electron distribution: $1s^2 2s^2 2p^4 3s^2$

Element C

Binding energy increases ←  → Binding energy decreases

Relative
intensity



Electron distribution: $1s^2 2s^2 2p^4 3s^2 3p^1$

Questions 7–11

7. Which element has the greatest atomic number? A. Element A B. Element B C. Element C D. Elements A and B E. Cannot be determined from spectra.
8. Which element has a valence-shell electron configuration of $3s^1$? A. Element A B. Element B C. Element C D. Elements A and C E. None.
9. Which statement best explains why binding energies generally increase from Element A to Element C? A. Increasing atomic radii B. Electrons occupy higher principal levels in C C. Nuclear charge increases while electrons remain in similar energy levels D. Fewer shielding electrons E. Different photon energies used.
10. Which element would be expected to have the largest first ionization energy? A. Element A B. Element B C. Element C D. Elements A and B E. Approximately the same.
11. Element C is exposed to photons of 1.20×10^3 kJ/mol. Which electrons would have the greatest kinetic energy after emission? A. 1s B. 2s C. 2p D. 3s E. 3p

Part III: Quantitative Analysis

A sample is irradiated with photons having a wavelength of 21.2 nm.

Use: $c = 3.00 \times 10^8 \text{ m/s}$; $h = 6.626 \times 10^{-34} \text{ J}\cdot\text{s}$; $N_A = 6.022 \times 10^{23} \text{ mol}^{-1}$; $1 \text{ eV} = 96.485 \text{ kJ/mol}$.

12. Calculate the energy of one photon in joules. Show your work.

13. Calculate the molar energy of the photons in kJ/mol. Show your work.

14. An electron is emitted from a subshell with a binding energy of 0.750 MJ/mol. Calculate the kinetic energy of the electron in kJ/mol. Show your work.

Part IV: AP-Style Free-Response Question

Photoelectron Spectroscopy of an Unknown Element

A student obtains a photoelectron spectrum for unknown element Y, using photons of energy $1.80 \times 10^3 \text{ kJ/mol}$. The spectrum contains five peaks:

Peak	Binding Energy (kJ/mol)	Relative Intensity
1	1,560	2
2	119	2
3	73.0	6
4	9.80	2
5	6.00	1

(a) Determine the total number of electrons in a neutral atom of element Y. Explain how relative intensities support your answer.

(b) Write the complete ground-state electron configuration for element Y.

(c) Identify the element represented by the spectrum.

(d) Identify the subshell represented by each peak (1–5).

(e) Calculate the kinetic energy of an electron emitted from Peak 5.

(f) Compare the binding energy of 3s electrons in Y vs. Mg; predict which is greater and justify.

(g) Evaluate the claim: Peak 3 represents a 3d subshell because its intensity is 6.

(h) Element Y forms a 3+ ion. Write the ion electron configuration and explain why electrons are removed from 3p/3s before 2p.

Answer Key

Multiple-Choice Answers

Q	Ans	Explanation
1	A	Highest binding energy = innermost 1s electrons.
2	E	Lowest binding energy = outermost 3p electrons.
3	C	$2+2+6+2+1 = 13$ electrons.
4	C	Subshell populations are $1s^2 2s^2 2p^3 3s^2 3p^1$.
5	C	Neutral atom with 13 electrons is aluminum.
6	B	2p electrons are closer and feel greater effective nuclear attraction.
7	C	Element C has 13 total electrons; A has 11; B has 12.
8	A	Element A has one electron in its outer 3s subshell.
9	C	Increasing nuclear charge increases attraction to electrons.
10	B	Element B (Mg) has higher first ionization energy than Element C (Al).
11	E	3p electrons have lowest binding energy, so highest KE after emission.

Quantitative Answers

12. $E = hc/\lambda = (6.626 \times 10^{-34} \text{ J}\cdot\text{s})(3.00 \times 10^8 \text{ m/s}) / (21.2 \times 10^{-9} \text{ m}) = 9.38 \times 10^{-18} \text{ J per photon.}$

13. $E_{\text{mol}} = (9.38 \times 10^{-18} \text{ J/photon})(6.022 \times 10^{23} \text{ photons/mol}) = 5.65 \times 10^6 \text{ J/mol} = 5.65 \times 10^3 \text{ kJ/mol.}$

14. $KE = h\nu - E_B = 5650 - 750 = 4900 \text{ kJ/mol.}$

Free-Response Scoring Guidelines

- (a) 2 points: 1 point for summing intensities ($2+2+6+2+1=13$); 1 point for stating intensity corresponds to electron count in each subshell.
- (b) 1 point: $1s^2 2s^2 2p^6 3s^2 3p^1$.
- (c) 1 point: Aluminum (Al).
- (d) 2 points: Peak 1→1s, Peak 2→2s, Peak 3→2p, Peak 4→3s, Peak 5→3p.
- (e) 1 point: $KE = 1800 - 6.00 = 1794 \text{ kJ/mol}$.
- (f) 2 points: 3s binding energy in aluminum is greater than in magnesium due to greater nuclear charge with similar shielding.
- (g) 1 point: Claim is incorrect; six electrons can indicate a p subshell, and peak position supports 2p, not 3d.
- (h) 2 points: $Y^{3+} = 1s^2 2s^2 2p^6$; electrons are removed first from 3p/3s because they have lower binding energies and are farther from the nucleus than 2p.